

SmartCare v0.3 - Domain Model Workbook

Week 6 student resource

Requirement-to-Concept Trace

Requirement	Concept	State/behaviour	Decision
Manage patients	Patient	Stores patient data	Include
Record patient name	Patient	Patient-name	Include
Manage practitioners	Practitioners	Stores practitioners' data	Include
Record practitioners name	Practitioners	Practitioners-name	Include
Manage appointments	Appointment	Stores appointment data	Include
Record appointment time	Appointment	Appointment -time	Include
Record a patient's identity	Record a patient's identity	Record a patient's identity	Record a patient's identity
Display appointment information	Display appointment information	Display appointment information	Display appointment information
Record a patient's identity	Record a patient's identity	Record a patient's identity	Record a patient's identity
Store appointment details	Store appointment details	Store appointment details	Store appointment details

CRC Cards

Patient

Responsibilities	Collaborators
Store patient name	Appointment
Maintain patient information	Appointment

Practitioner

Responsibilities	Collaborators
Store practitioner name	Appointment
Maintain practitioner information	Appointment

Appointment

Responsibilities	Collaborators
Store appointment time	Patient
Link a patient to an appointment	Practitioner

Optional class

Responsibilities	Collaborators

UML Class Diagram

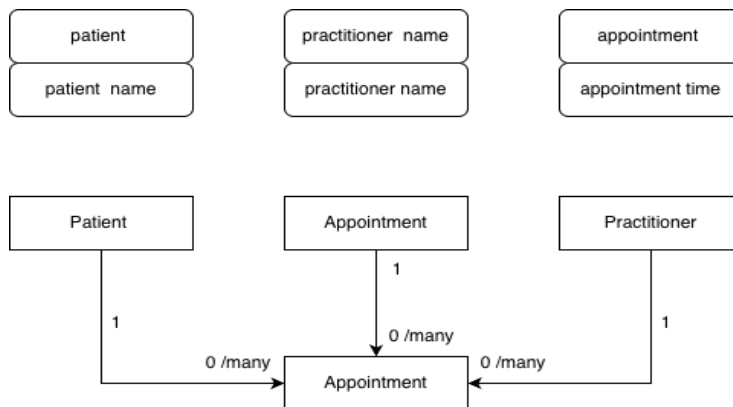
Insert/draw UML here. Include defensible relationships and multiplicities.

One Patient can have zero or many Appointments.

One Practitioner can have zero or many Appointments.

1 = exactly one

0/many = zero or many



Design Rationale

Explain class selection, responsibility allocation and key relationships.

Selection: The classes in the UML are **Patient**, **Practitioner** and **Appointment**. I selected them because these are the main identified concepts in the SmartCare requirements. The system needs work on the management of patients, practitioners and appointments.

Responsibility allocation: The patient class is responsible for all data about the patient. The practitioner class is responsible for all strong data about the practitioner. And appointment class is responsible for all appointment time and schedule.

Key relationships: Initially patients are associated with appointments because it is made for the patients. A patient can have zero or many appointments. Here one appointment is associated with one patient. Again a practitioner can have zero or many appointment. Here one appointment is associated with one practitioner.

AI Design Review Record

AI suggestion	Evidence	Decision	Reason	Model change
PatientManager	Requirement says SmartCare needs to manage patients.	Reject	The requirement supports Patient , but does not require a separate manager class.	No change
PractitionerManager	Requirement says SmartCare needs to manage practitioners.	Reject	The requirement supports Practitioner , but not a separate manager.	No change
AppointmentManager	Requirement says	Reject	Appointment is already the	No change

	SmartCare needs to manage appointments.		required domain concept.	
ScheduleEngine	No confirmed requirement for an automated scheduling engine.	Reject	It adds functionality that has not been requested.	No change
Patient-Appointment association	Appointments record patient information.	Accept	The relationship is directly supported by the appointment requirement.	Added association